

Rojae Mighty

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Research Profile

Physics undergraduate researcher exploring quantitative finance, machine learning, and data-driven modeling alongside nuclear physics, gluon saturation, geometric scaling, and computational methods for high-energy collisions. His work combines Python, numerical analysis, C++/ROOT, and computational modeling.

Education

Stony Brook University, Stony Brook, NY | **B.S. in Physics**, Expected May 2028 | GPA: 3.7

Relevant Coursework: Electromagnetic Theory, Statistical Mechanics, Computational Methods, Calculus, Linear Algebra

Research Experience

Brookhaven National Laboratory - Upton, NY

DOE Science Undergraduate Laboratory Internship Researcher | June 2025-August 2025; June 2026-August 2026

- Developed ROOT, Python, and Linux scripts to support electron-ion collision research on 10 million simulated events.
- Built reproducible analysis workflows for studying particle behavior and nuclear effects after high-energy collisions.
- Implemented and tested analysis code for semi-inclusive electron-ion collision studies using spatially dependent nuclear structure inputs.
- Worked independently within an 8-person research team, translating physics requirements into validated code and organized outputs.

Center for Frontiers in Nuclear Science (CFNS) - Stony Brook, NY

Undergraduate Researcher, Nuclear and High-Energy Physics | January 2025-Present

Advisor: Zhoudunming (Kong) Tu

- Built Python and scikit-learn pipelines to reconstruct key kinematic quantities from 10 million particle-collision events.
- Used C++, ROOT, Monte Carlo event generators, and numerical simulations to study gluon saturation and geometric scaling at the Electron-Ion Collider.
- Developed machine-learning approaches for Deep Inelastic Scattering kinematic reconstruction, including extraction of x and Q^2 for inclusive and extended scattering channels.
- Created reusable workflows that improved repeatability, verification, and sharing of analysis results across research tasks.

Presentations, Awards, and Research Recognition

- **Geometric Scaling and Gluon Saturation at the EIC** - Deep Inelastic Scattering Conference, Bologna, Italy, 2026; APS Global Physics Summit, 2026; Yale Undergraduate Research Conference, 2026; NSBP Annual Conference, 2025; APS DNP Meeting, 2025
- 1st Place for Physical Sciences Presentation, Yale Undergraduate Research Conference, January 2026
- Outstanding Oral Presentation Award, National Society of Black Physicists Conference, January 2025
- Barry Barish Undergraduate Research Travel Award, Stony Brook University, January 2026
- Edward Guiliano Global Fellowship, Stony Brook University, January 2026

Leadership and Service

Dean's Student Leadership and Advisory Council, College of Arts and Sciences

2024-Present

- Represent undergraduates to the CAS Dean on policy, student experience, and campus services.
- Serve as a CAS Ambassador at admissions events and the Godfrey Excellence in Teaching Awards.

Collegiate Science and Technology Entry Program, Stony Brook University

2024-Present

- Mentor underrepresented STEM undergraduates on research pathways, academic planning, and professional development.

Technical Skills

Programming: Python, C++, C, Unix/Linux shell scripting

Research Tools: ROOT, Monte Carlo event generators, numerical simulation, Git, GitHub, LaTeX, Linux/Unix environments

Data and ML: pandas, NumPy, scikit-learn, feature engineering, statistical analysis

Domains: Electron-Ion Collider physics, particle-collision data, gluon saturation, geometric scaling, DIS kinematics